



James Clerk Maxwell
(1831-1879)
Scottish physicist

Maxwell, widely considered to have been one of the most significant figures of the nineteenth century.

His work fundamentally changed conceptions of electromagnetism and introduced the basis of field theory. He is also known for his work on thermodynamics and the kinetic theory of gases.

Chapter 12 Maxwell 's Equations; Magnetism of Matter

12.1 Displacement current and induced magnetic field

12.2 Maxwell 's Equations

12.3 Magnetic materials

12.1 Displacement current and induced magnetic field

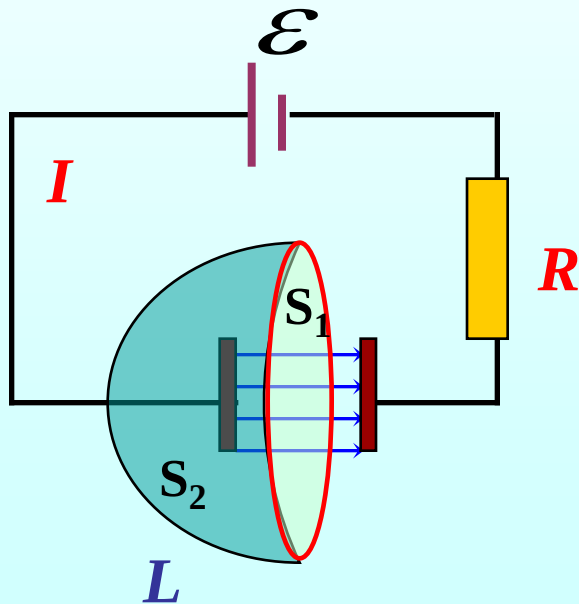
1. Displacement current (位移电流)

$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 \sum_i I_i$$

Ampère's law

$$I = \int_s \vec{j} \cdot d\vec{A}$$

$\vec{j}$: current density

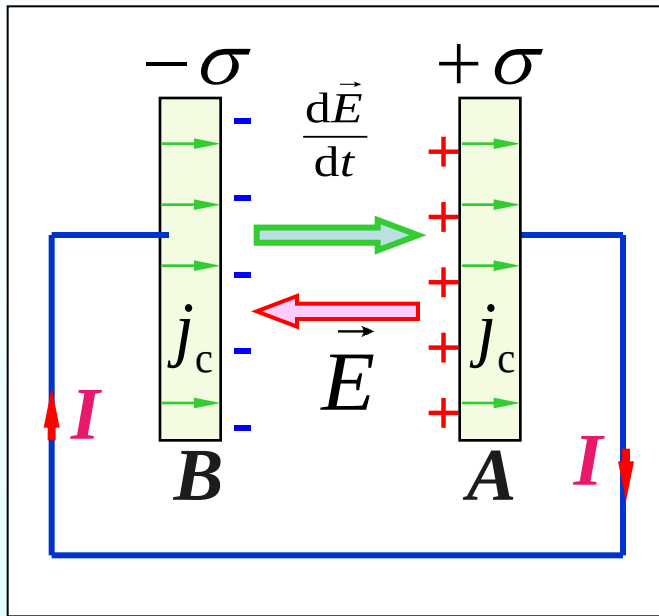


Different surfaces bounded by the same closed path

$$\oint_L \vec{B} \cdot d\vec{l} = \int_{S_1} \vec{j} \cdot d\vec{A} = 0$$

$$\oint_L \vec{B} \cdot d\vec{l} = \int_{S_2} \vec{j} \cdot d\vec{A} = \mu_0 I$$

Ampère's law fails because no current is flowing through surface 1.



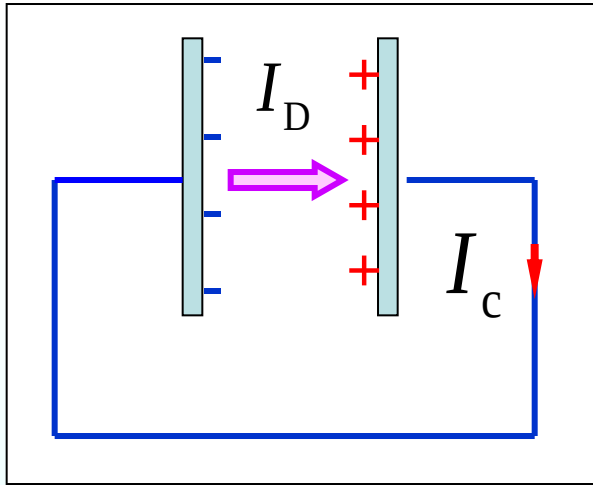
$$I_c = \frac{dQ}{dt}$$

$$Q = CV = \left(\epsilon_0 \frac{A}{d}\right)(Ed) = \epsilon_0 AE$$

$$I_c = \frac{dQ}{dt} = \epsilon_0 A \frac{dE}{dt} = \epsilon_0 \frac{d\Phi_E}{dt}$$

Discharging course

Maxwell thought there needs to be an extra term in **Ampère's** law involving the **changing electric field**.



$$I_d = \epsilon_0 \frac{d\Phi_E}{dt} = I_c = \frac{dQ}{dt}$$

Maxwell called it a **displacement current**

$$j_d = \frac{I_d}{A} = \epsilon_0 \frac{d\Phi_E}{A dt} = \epsilon_0 \frac{dE}{dt}$$

displacement current density

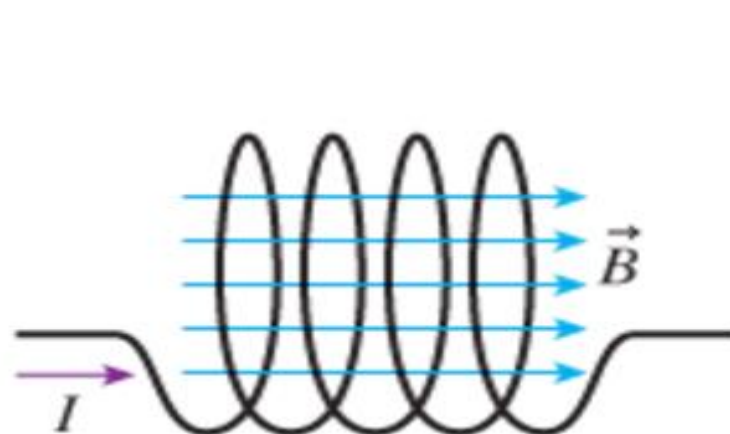
$$I_s = I_c + I_d$$

general form of a current 全电流

Current concept includes conduction current and displacement current and current is continuous.

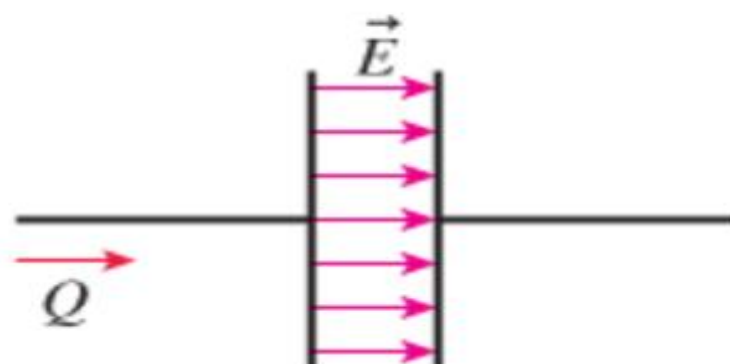
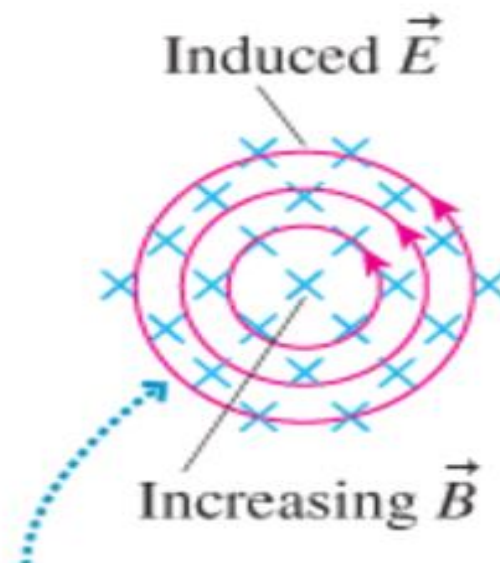
$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 I = \mu_0 (I_c + I_d) = \mu_0 I_c + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} \quad \text{General form}$$

Maxwell :A changing electric field produces a displacement current, and it is equivalent to an ordinary electric current.



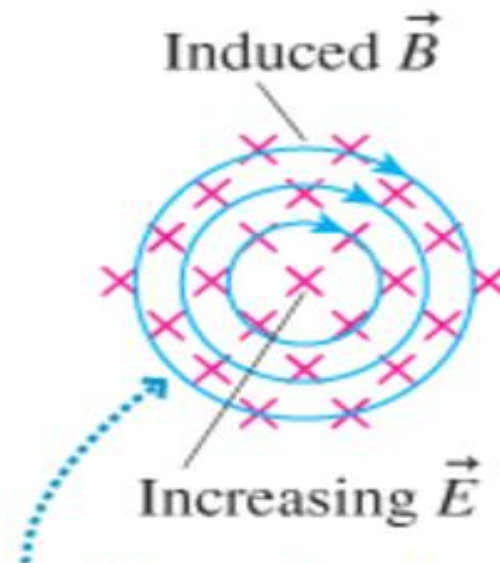
Increasing solenoid current

Faraday's law describes an induced electric field.



Increasing capacitor charge

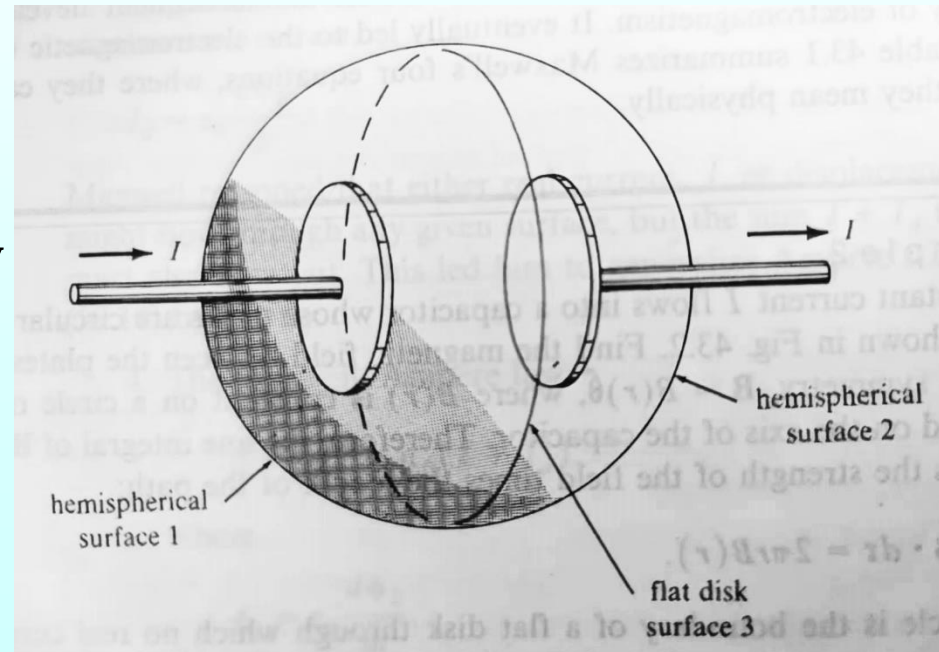
The Ampère-Maxwell law describes an induced magnetic field.



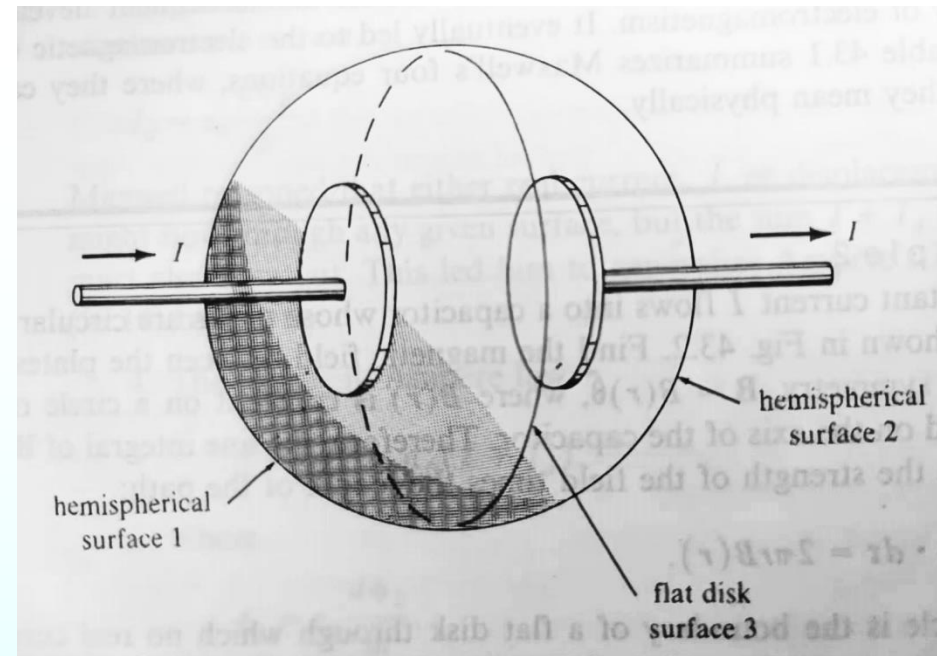
A circular parallel plate capacitor of **radius R** is **being charged** by a constant **current I** . The capacitor is contained within an imaginary spherical surface that is divided into two hemispheres (surfaces 1 and 2) and a flat circular disk (surface 3) that passes between plates of capacitor. Determine the **electric flux**, the **real current** and **displacement current** for each of following:



1. Surface 1
2. Surface 2
3. Surface 3
4. The spherical region bounded by surfaces 1 and 2
5. The hemispherical region bounded by surfaces 1 and 3.



- 1.Surface 1
- 2.Surface 2
- 3.Surface 3
4. The spherical region bounded by surfaces 1 and 2
- 5.The hemispherical region bounded by surfaces 1 and 3.



$$(1) \Phi_E = 0, \quad I_c = -I, \quad I_d = 0$$

$$(2) \Phi_E = 0, \quad I_c = I, \quad I_d = 0$$

$$(3) \Phi_E = It / \epsilon_0, \quad I_c = 0, \quad I_d = I$$

$$(4) \Phi_E = 0, \quad I_c = 0, \quad I_d = 0$$

$$(5) \Phi_E = It / \epsilon_0, \quad I_c = -I, \quad I_d = I$$

$$I_c = \frac{dq}{dt} = \int_s \vec{j} \cdot d\vec{A}$$

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt} = \epsilon_0 \frac{d \int_s \vec{E} \cdot d\vec{A}}{dt}$$

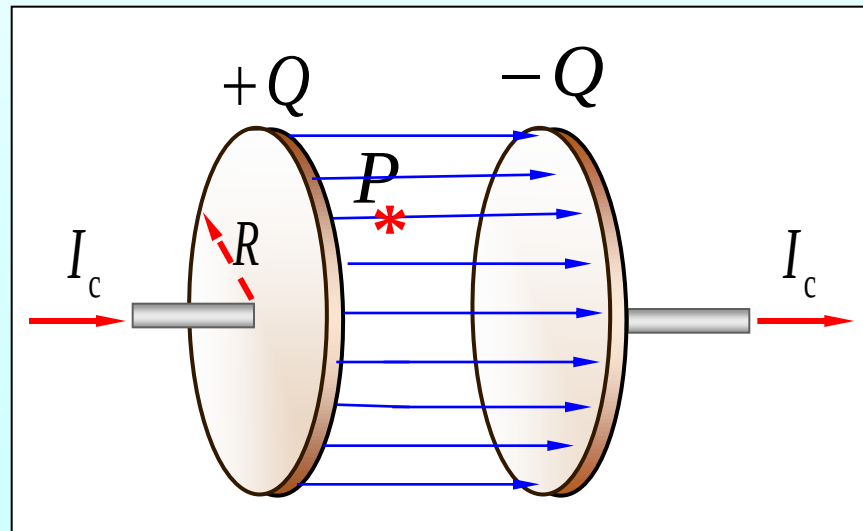
Example : Charging capacitor

A air-gap capacitor has circular plates of radius $R=3.0\text{cm}$ when the plate is changing by $I_c = \frac{dQ}{dt} = 2.5\text{A}$

Determine:

(a) The displacement current I_d ;

(b) The magnetic field induced at point a distance 2.0cm from the center.



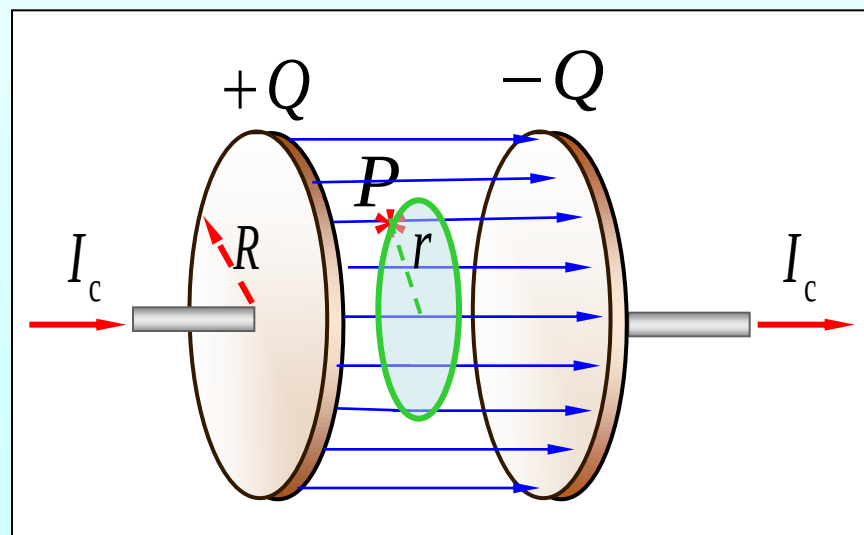
Solution: choose the circular path with radius r

$$\Phi_E = \pi r^2 E = \pi \mathbf{r}^2 \frac{\sigma}{\epsilon_0} = \frac{\pi r^2}{\pi R^2} \frac{Q}{\epsilon_0} = \frac{Q}{\epsilon_0} \frac{r^2}{R^2}$$

$$I_d(r) = \epsilon_0 \frac{d\Phi_E}{dt}$$

$$= \epsilon_0 \frac{1}{\epsilon_0} \frac{r^2}{R^2} \frac{dQ}{dt} = \frac{dQ}{dt} \frac{r^2}{R^2}$$

$$= I_c \frac{r^2}{R^2}$$



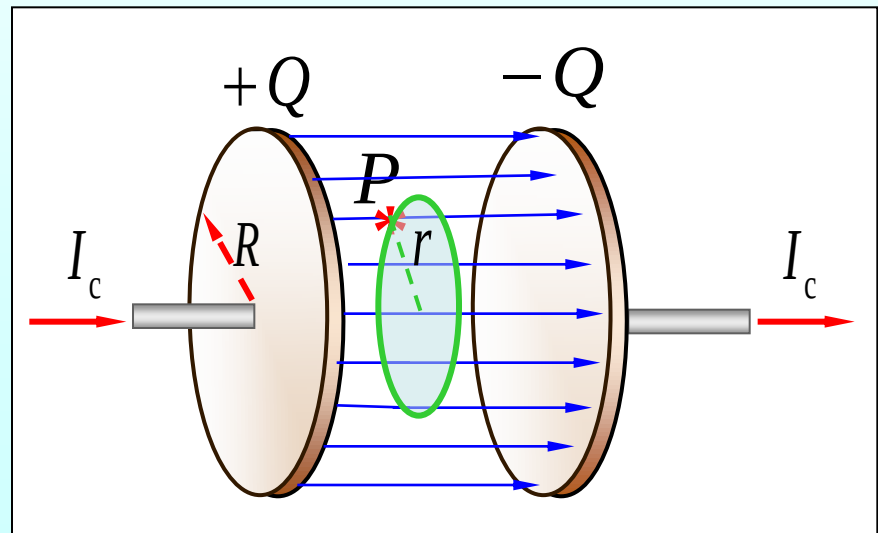
Apply **Ampère's law** (with displacement current)

$$\therefore \oint_l \vec{B} \cdot d\vec{l} = \mu_0 (I_c + I_d) = \mu_0 I_d$$

$$B(2 \pi r) = \mu_0 \frac{r^2}{R^2} \frac{dQ}{dt}$$

$$\begin{aligned} \therefore B &= \frac{\mu_0 r}{2 \pi R^2} \frac{dQ}{dt} \\ &= \frac{\mu_0 r}{2 \pi R^2} I_c \end{aligned}$$

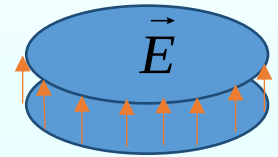
$$r > R \quad B = \frac{\mu_0}{2 \pi r} I_c$$



In Fig. an electric field is directed out of the page within a circular region of radius $R=3.00$ cm. The magnitude of the electric field is given by $E = (0.500\text{V} / \text{m} \cdot \text{s})(1 - r / R)t$, where t is in seconds and r is the radial distance $r \leq R$. What is the magnitude of the **induced magnetic field** at radial distances (a) 2.00 cm and (b) 5.00 cm?

Solution: (a) the enclosed electric flux is for $r \leq R$

$$\Phi_E = \int_0^r E \, 2\pi r \, dr = t(0.500)(2\pi) \int_0^r \left(1 - \frac{r}{R}\right) r \, dr = t\pi \left(\frac{1}{2} r^2 - \frac{r^3}{3R} \right)$$



$$B(2\pi r) = \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 \mu_0 \pi \left(\frac{1}{2} r^2 - \frac{r^3}{3R} \right) \quad B = 3.09 \times 10^{-20} \text{ T.}$$

(b) the enclosed electric flux is $r > R$

$$\Phi_E = \int_0^R E \, 2\pi r \, dr = t(0.500)(2\pi) \int_0^R \left(1 - \frac{r}{R}\right) r \, dr = t\pi \left(\frac{1}{2} R^2 - \frac{R^3}{3R} \right) = \frac{1}{6} t\pi R^2$$

$$B = \frac{\varepsilon_0 \mu_0 R^2}{12r} = \frac{(8.85 \times 10^{-12})(4\pi \times 10^{-7})(0.030)^2}{12(0.0500)} = 1.67 \times 10^{-20} \text{ T.}$$

12.2 Maxwell's Equations

1. Electric Field

(1) Faraday's law (circuital theorem) 环路定理

Electrostatic field

$$\oint_L \vec{E} \cdot d\vec{l} = 0$$

Induced electric field

$$\oint_L \vec{E} \cdot d\vec{l} = -\frac{d\Phi_m}{dt} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{A}$$

$$\oint_L \vec{E} \cdot d\vec{l} = -\frac{d\Phi_m}{dt} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{A}$$

A changing magnetic field (also) produce a electric field.

(2) Gauss's law

Electrostatic field

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{encl}}{\epsilon_0}$$

Induced electric field

$$\oint_S \vec{E} \cdot d\vec{A} = 0$$

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{encl}}{\epsilon_0}$$

The electric flux through any closed surface is equal to the net charge enclosed by the surface divided by permittivity of free space ϵ_0 .

Charges create electric fields.

2 . Magnetic Field

(1) Ampère 's law

$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 (I_c + I_d) = \mu_0 I_c + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

Magnetic fields can be created either by **currents** or by **changing electric fields**.

A changing electric field produce a magnetic field.

(2) Gauss's law for magnetism

$$\oint_S \vec{B} \cdot d\vec{A} = 0$$

The net magnetic flux through any closed surface is equal to zero. *There are no isolated magnetic poles.*

Maxwell 's Equations

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{encl}}{\epsilon_0} + 0 = \frac{Q_{encl}}{\epsilon_0}$$

$$\oint_L \vec{E} \cdot d\vec{l} = 0 + \left(-\frac{d\Phi_m}{dt}\right) = -\frac{d\Phi_m}{dt} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{A}$$

$$\oint_S \vec{B} \cdot d\vec{A} = 0 + 0 = 0$$

$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 I_c + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

Maxwell's equations for the fields, together with the Lorentz force law to tell us how matter responds to the fields, form the complete theory of electromagnetism.

The Lorentz equation

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

Maxwell 's Equations

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{encl}}{\epsilon_0} = \frac{\int_V \rho dV}{\epsilon_0}$$

$$\nabla \cdot \vec{E} = \rho_e / \epsilon$$

$$\nabla \cdot \vec{B} = 0$$

$$\oint_L \vec{E} \cdot d\vec{l} = -\frac{d\Phi_m}{dt} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{A}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\oint_S \vec{B} \cdot d\vec{A} = 0$$

$$\nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 I_c + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} = \mu_0 \left[\int_S (\vec{j}_c + \epsilon_0 \frac{\partial \vec{E}}{\partial t}) \cdot d\vec{A} \right]$$

Without isolated magnetic poles

The Lorentz equation

$$\vec{F} = q_e (\vec{E} + \vec{v} \times \vec{B})$$

Maxwell 's Equations

$$\nabla \cdot \vec{E} = \rho_e / \epsilon_0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{j}_e + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

With isolated magnetic poles

The Lorentz equation

$$\vec{F} = q_e (\vec{E} + \vec{v} \times \vec{B}) + q_m (\vec{B} - \vec{v} \times \vec{E} / c^2)$$

Maxwell 's Equations

$$\nabla \cdot \vec{E} = \rho_e / \epsilon_0$$

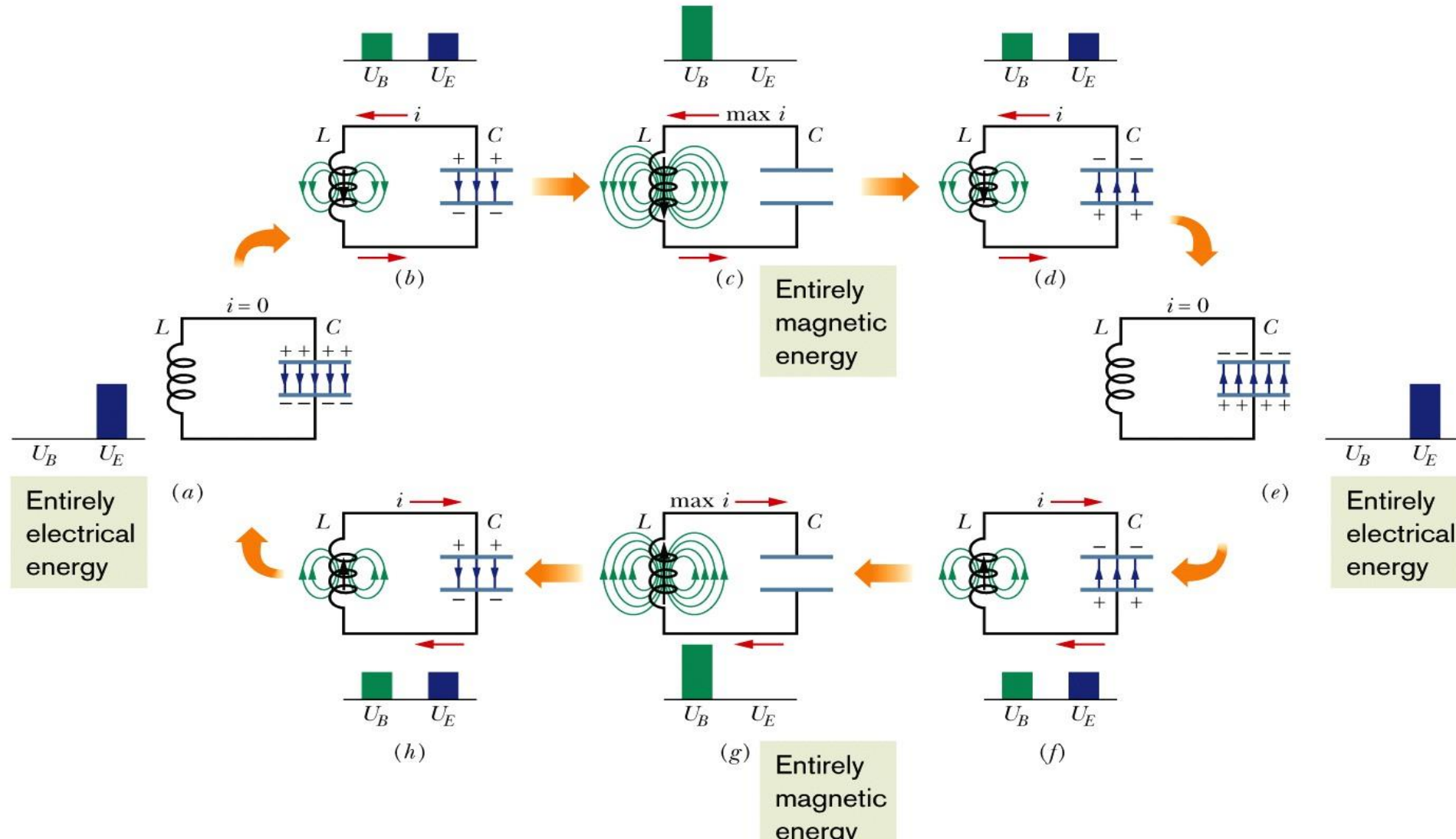
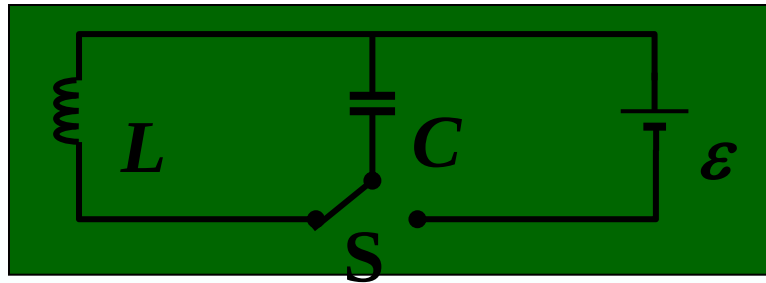
$$\nabla \cdot \vec{B} = \mu_0 \rho_m$$

$$\nabla \times \vec{E} = -\mu_0 \vec{j}_m - \frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{j}_e + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

* *LC* oscillator :

$$\omega = \sqrt{\frac{1}{LC}}$$



* The total energy U present at any instant in an oscillating LC circuit is given by

$$U = U_B + U_E = \frac{Li^2}{2} + \frac{q^2}{2C}$$

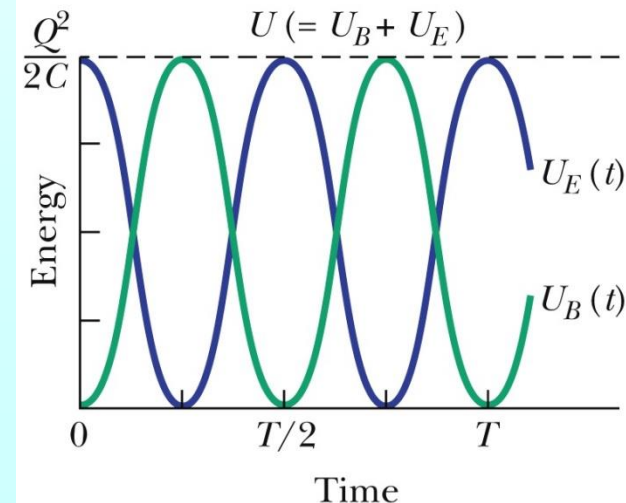
$$\frac{dU}{dt} = \frac{d}{dt} \left(\frac{Li^2}{2} + \frac{q^2}{2C} \right) = Li \frac{di}{dt} + \frac{q}{C} \frac{dq}{dt} = 0.$$

$$i = \frac{dq}{dt} \text{ and } \frac{di}{dt} = \frac{d^2q}{dt^2}.$$

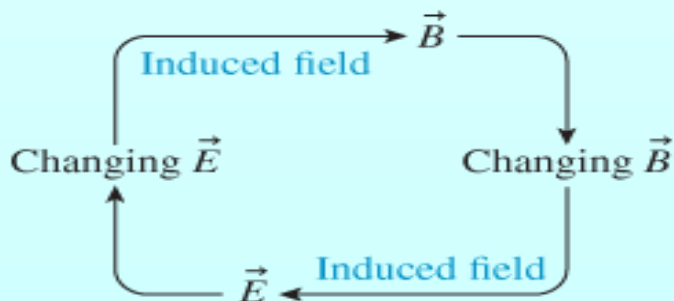
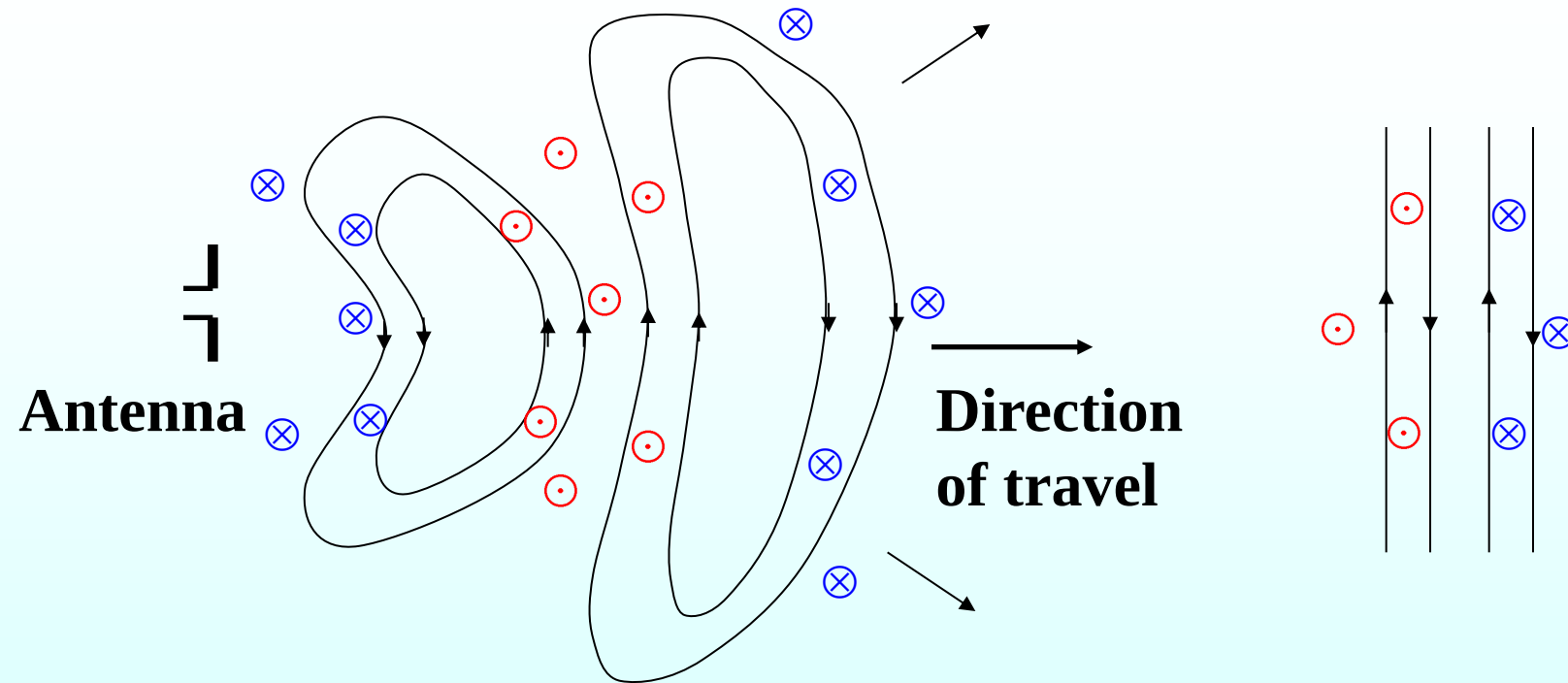
$$\frac{d^2q}{dt^2} + \frac{1}{LC} q = 0$$

$$\omega = \frac{1}{\sqrt{LC}} \quad (LC \text{ circuit}).$$

The electrical and magnetic energies vary but the total is constant.

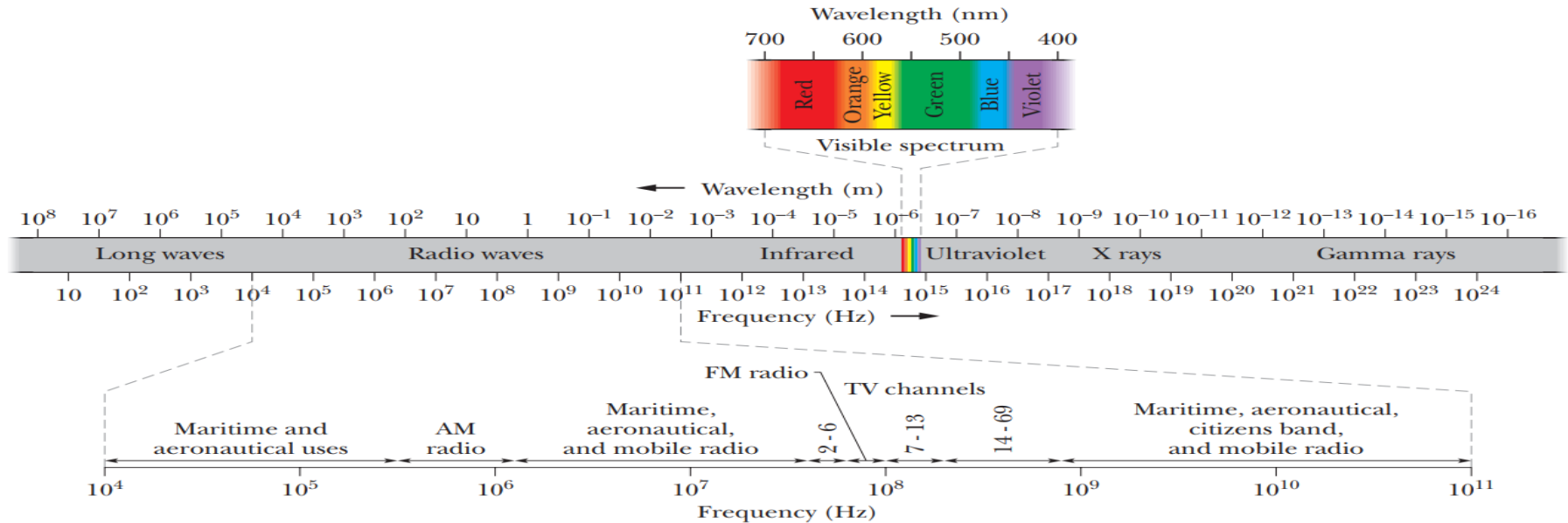


*Electromagnetic waves

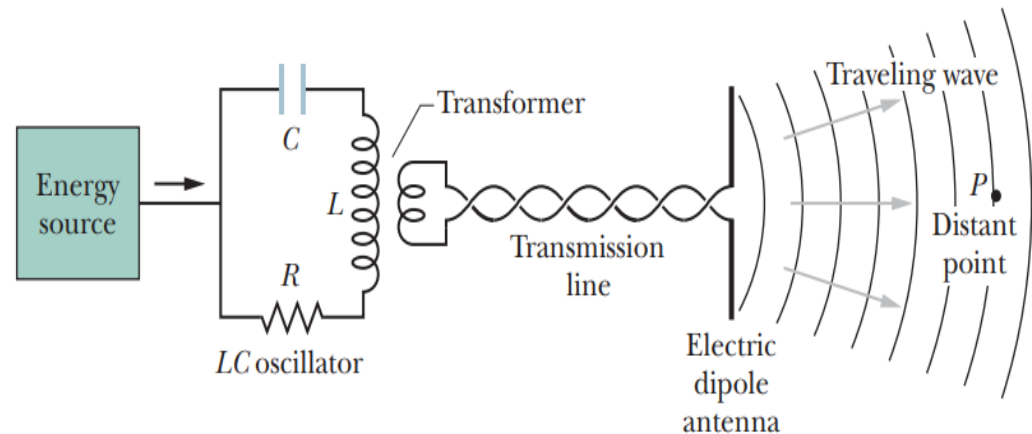


Accelerating electric charges give rise to electromagnetic waves.

Maxwell also calculated that the **speed** of propagation of an electromagnetic field is approximately that of the **speed of light**. He proposed that the phenomenon of light is therefore an electromagnetic phenomenon.

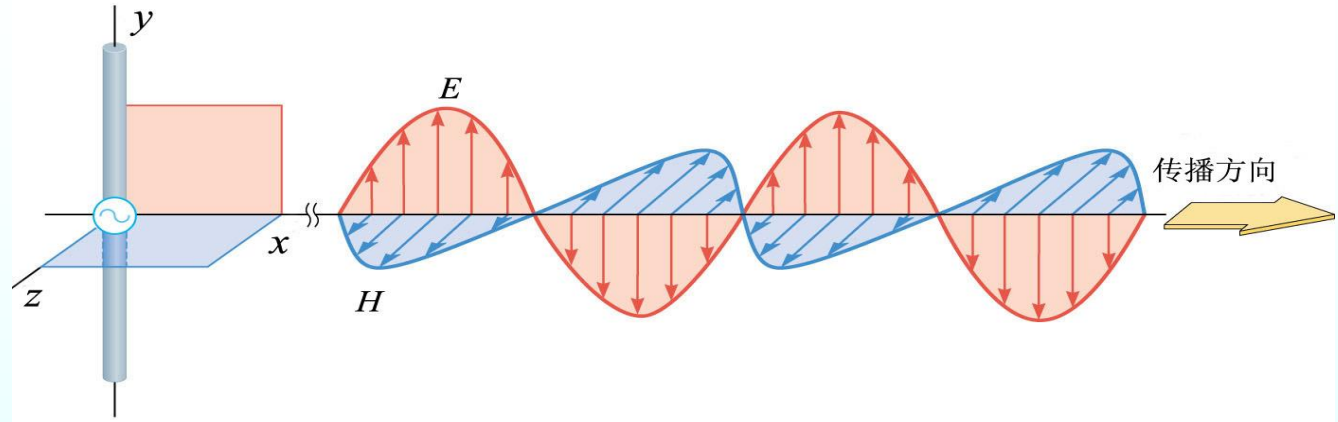


This achievement opened a whole new world of communication-first the wireless telegraph, then radio and television.



◆ Speed in free space

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$



Wave equation

$$\left\{ \begin{array}{l} E = E_0 \cos \omega \left(t - \frac{r}{v} \right) \\ B = B_0 \cos \omega \left(t - \frac{r}{v} \right) \end{array} \right.$$

- The electric and magnetic fields at any point are perpendicular to each other and to the direction of motion.
- The electric and magnetic fields are “in phase”.

Classical physics

Newton's first law

Newton's second law

Newton's third law

Newton's law of gravity

Gauss's law

Gauss's law for magnetism

Faraday's law

Ampère-Maxwell law

Lorentz force law

First law of thermodynamics

Second law of thermodynamics

Key words And Terms

magnetization

磁化

magnetization curve

磁化曲线

hysteresis loop

磁滞回线

retentivity

剩磁

demagnetization

退磁

displacement current

位移电流

12.3 Magnetic Materials

1. Materials in magnetic field

All materials are magnetic to some extent.

$$\vec{B} = \vec{B}_0 + \vec{B}'$$

与介质有关的电流产生

(1) Paramagnetic (顺磁质) $\vec{B}' \uparrow\uparrow \vec{B}_0, \vec{B} > \vec{B}_0$

(铝、氧、锰等)

(2) Diamagnetic (抗磁质) $\vec{B}' \uparrow\downarrow \vec{B}_0, \vec{B} < \vec{B}_0$

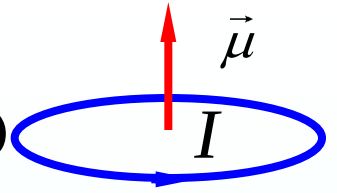
(铜、铋、氢等)

(3) Ferromagnetic (铁磁质) $\vec{B}' \uparrow\uparrow \vec{B}_0, \vec{B} \gg \vec{B}_0$

(铁、钴、镍等)

2. Mechanism of magnetizing materials

(1) Magnetic moment of molecules (分子磁矩)



Atoms consists of electrons and atomic nucleus .

(a) An electron orbits around the nucleus.

(b) Electrons have an intrinsic angular momentum called spin.

Orbital angular momentum $\Rightarrow$ current loop
 $\Rightarrow$ **orbital magnetic dipole moment**

Spin angular momentum $\Rightarrow$ current loop
 $\Rightarrow$ **Spin magnetic dipole moment**

Electron magnetic dipole moments $\Rightarrow$

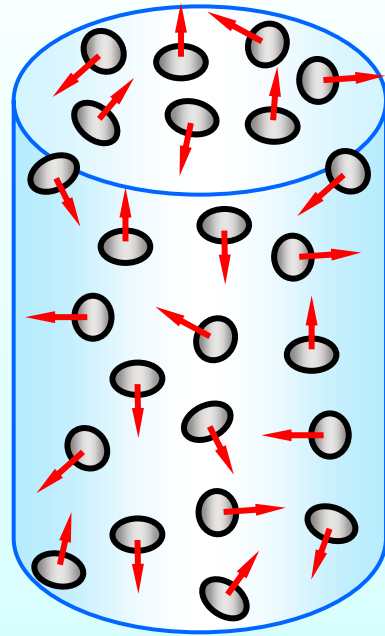
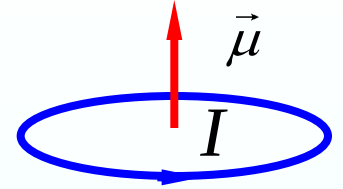
Atom magnetic dipole moments $\Rightarrow$

Molecular magnetic dipole moments

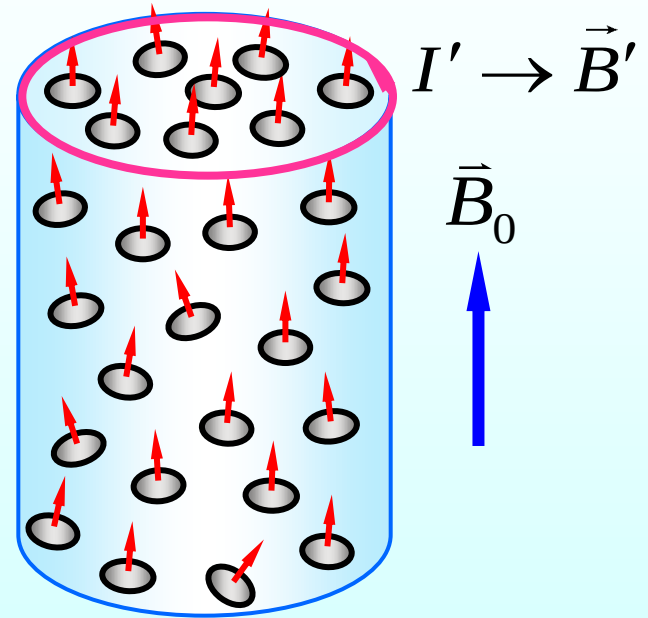
(2) Paramagnetic materials

Molecules have a **permanent magnetic dipole moment**.

No external field: randomly orient $\vec{\mu} \neq 0$



No external field



In external field

Applied external field tend to align the molecular magnetic moment.

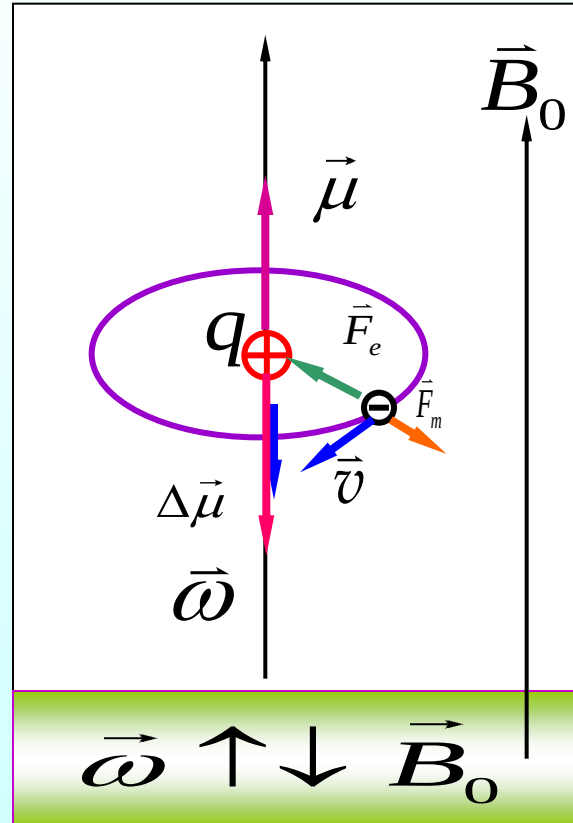
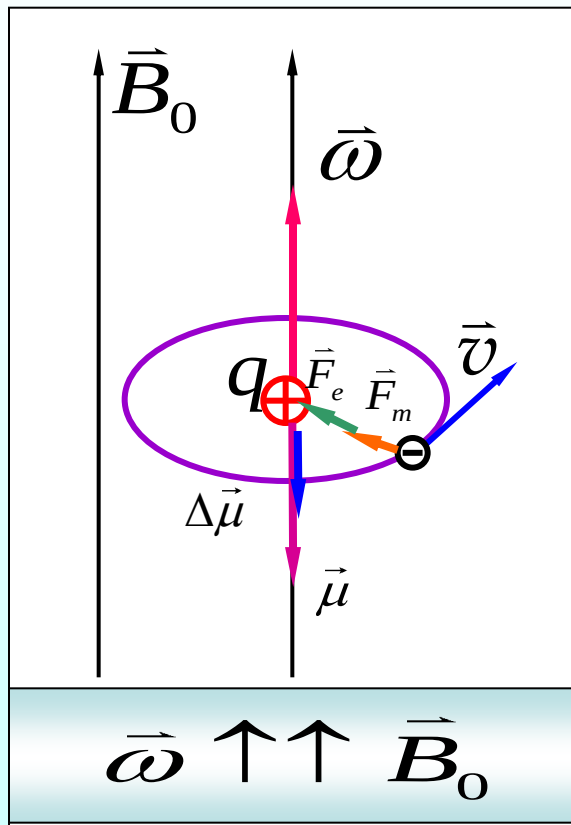
The total magnetic field $B = B_0 + B'$ I' 束缚电流

(3) Diamagnetic materials

Molecules have no permanent magnetic dipole moment.

No external field $\vec{\mu} = 0$

In external field

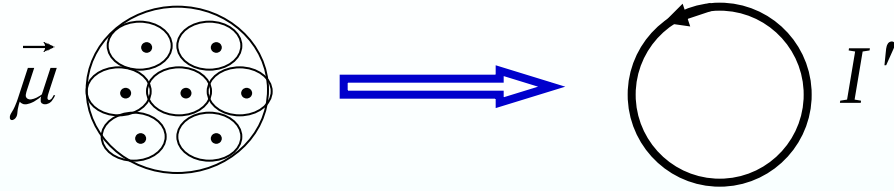


The total magnetic field

$$B = B_0 - B'$$

Everything is slightly diamagnetic.

3. 介质中的安培环路定理 补充



$$\oint_L \vec{B} \cdot d\vec{l} = \mu \sum_i I_i$$

$$\mu = K_m \mu_0$$

μ Magnetic permeability of a material

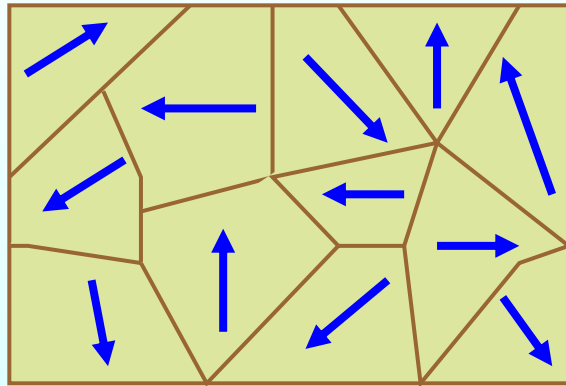
K_m Relative permeability

注意： 对 I 的求和并不包括束缚电流，因为束缚电流对磁场的贡献已经包含在 μ 中。

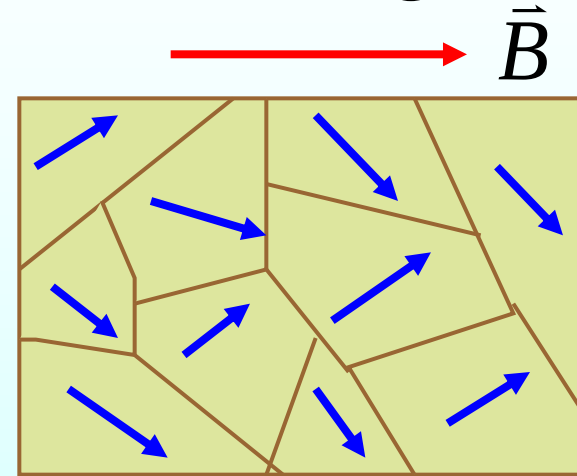
4. Ferromagnetism

Magnetic domains 磁畴

No external magnetic field



In external magnetic field



Curie temperature: ferromagnetic $\rightarrow$ paramagnetic

In external magnetic field magnetization of domains may rotate or the borders of domains move so as to the magnetic orientation of domains is parallel to the external field.

Segment ab: magnetization (磁化曲线)

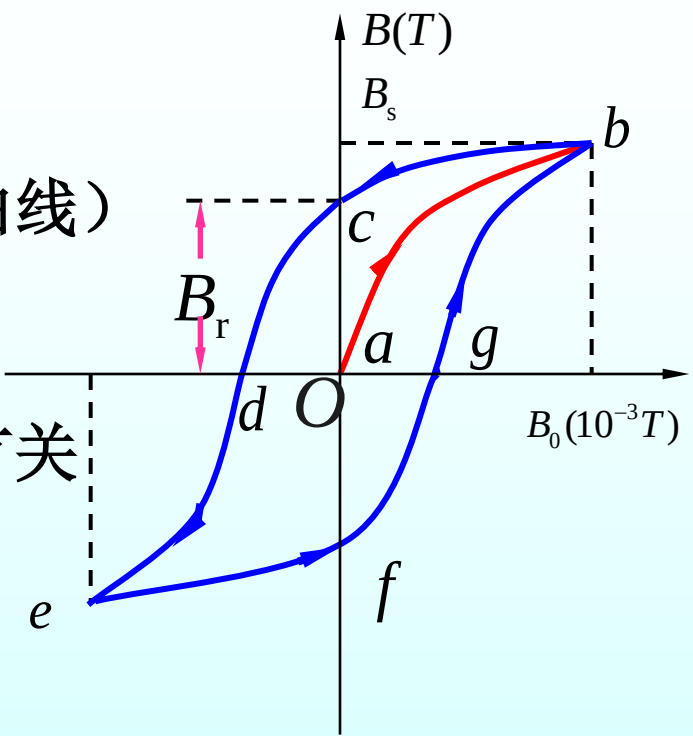
B_s :saturation (饱和磁感应强度)

B_r :retentivity (剩磁)

Segment cd: demagnetization(退磁曲线)

特点:

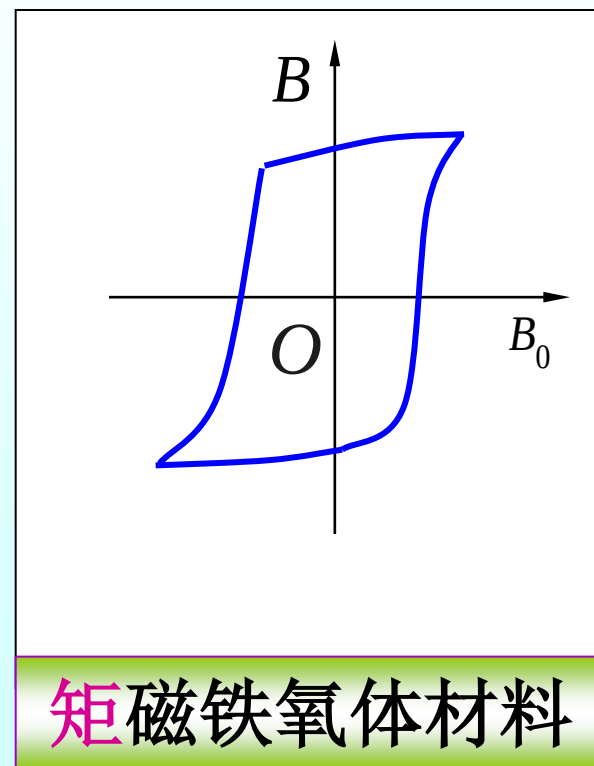
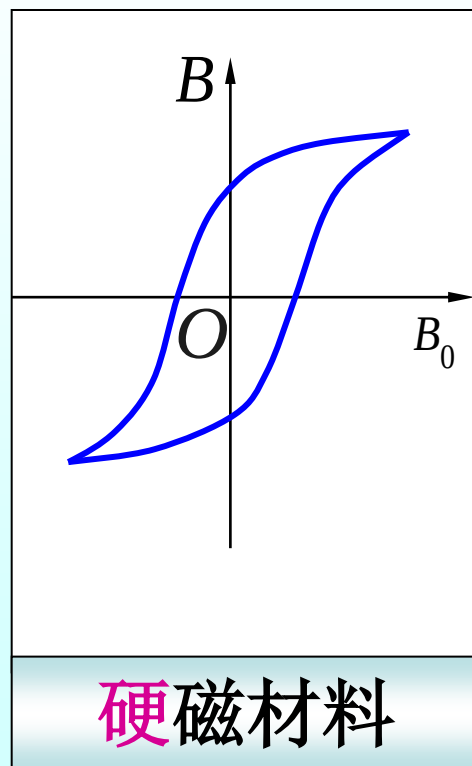
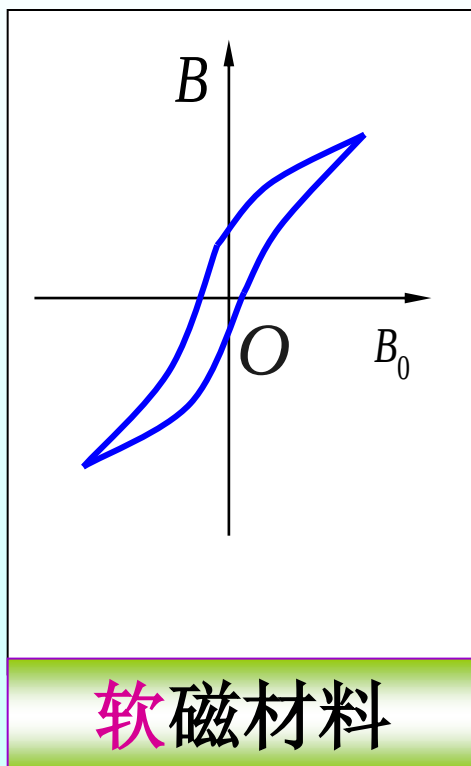
- 1. K_m 非常大，且不是常数，与外场有关
- 2. 磁滞现象，不可逆过程.
 $\vec{B}$ 变化落后于 $\vec{B}_0$ 变化
- 3. 回线包围的面积，反映能量消耗。



Hysteresis loop

Curie temperature: If the temperature of a ferromagnetic material is raised above a certain critical value, most such materials then become simple paramagnetic.

* Ferromagnetic materials

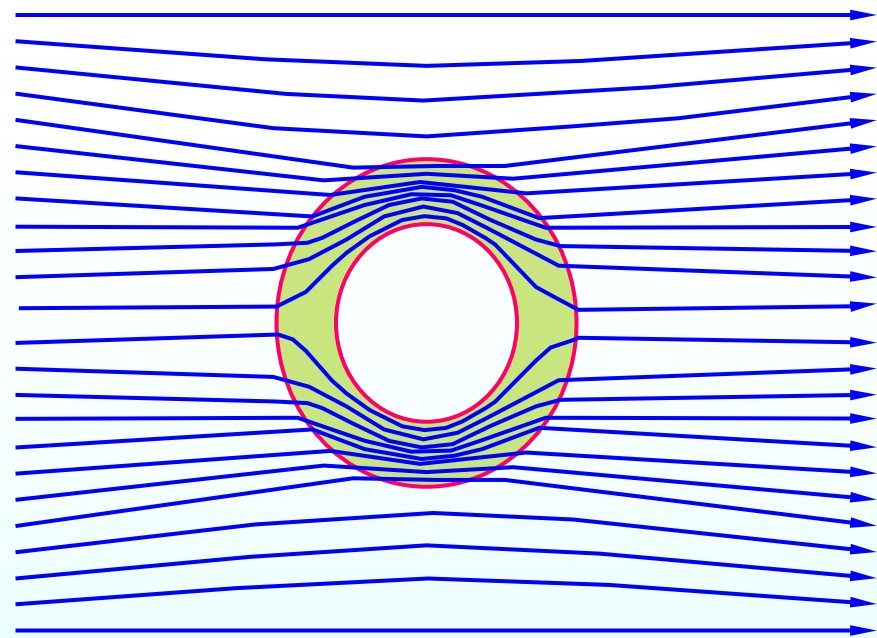


静磁屏蔽

把磁导率不同的两种磁介质放到磁场中，在它们的交界面上磁场要发生突变，引起了磁感应线的折射。

“静磁真空”

超导体迈斯纳效应：超导体是完全的抗磁体，内部的磁感应强度为零，超导体排斥体内的磁场可以把全部磁通量排出。



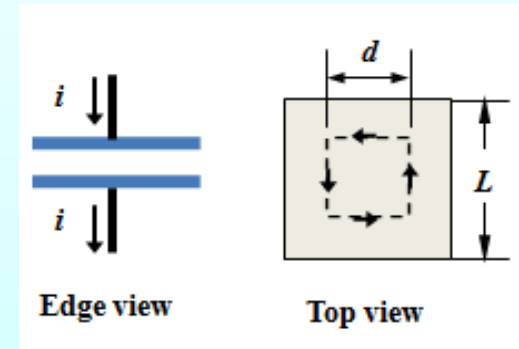
In fig. a parallel plate capacitor has square plates of the edge length $L=1.0\text{m}$. A current of 2.0A charges the capacitor producing a uniform electric field $\vec{E}$ between the plates, with $\vec{E}$ perpendicular to the plates. (a) What is the displacement current I_d through the region between the plates? (b) What is dE/dt in the region? (c) What is the displacement current encircled by the square dashed path of edge length $d=0.50\text{m}$? (d) What is $\oint \vec{B} \cdot d\vec{l}$ around this square dashed path?

(a) At any instant the displacement current I_d in the gap between the plates equals the conduction current I_c in the wires. Thus $I_d = I_c = 2.0\text{ A}$.

$$(b) \frac{dE}{dt} = \frac{1}{A} \left(\frac{d\Phi_E}{dt} \right) = \frac{1}{\epsilon_0 A} (\epsilon_0 \frac{d\Phi_E}{dt})$$

$$= \frac{I_d}{\epsilon_0 A} = \frac{2.0\text{ A}}{(8.85 \times 10^{-12} \text{ F/m})(1.0\text{m})^2} = 2.3 \times 10^{11} \text{ V} \cdot \text{m/s}$$

$$(c) I'_d = I_d \left(\frac{d^2}{L^2} \right) = (2.0\text{ A}) \left(\frac{0.50\text{m}}{1.0\text{m}} \right)^2 = 0.50\text{ A}$$



$$(d) \oint_L \vec{B} \cdot d\vec{l} = \mu_0 I'_d = (4\pi \times 10^{-7} \text{ H/m})(0.50\text{ A}) = 6.3 \times 10^{-7} \text{ T} \cdot \text{m}$$

Suppose an air-gap capacitor has circular plates of radius $R=2.5\text{cm}$ and separation $d=2.0\text{mm}$. A 96.0-Hz emf (电动势), $\mathcal{E} = \mathcal{E}_0 \cos \omega t$, is applied to the capacitor. The maximum displacement current (位移电流) is $35\mu\text{A}$. Determine (a) the maximum conduction current (传导电流) I_c , (b) the value of $\mathcal{E}_0$ (电动势), (c) the maximum value of $d\Phi_E/dt$ between the plates. Neglect fringing. (忽略边缘效应, 即平板间可视为匀强电场)

$$(a) I_{c,\max} = I_{d,\max} = 35\mu\text{A}$$

$$(b) I_c = \frac{dQ}{dt} = \frac{C\mathcal{E}(\text{emf})}{dt} = -C\mathcal{E}_0\omega \sin \omega t$$

$$\mathcal{E}_0 = \frac{I_{c,\max}}{\omega C} = \frac{I_{c,\max}}{\omega \mathcal{E}_0 \frac{A}{d}} = 6.68 \times 10^3 \text{ V}$$

$$(c) I_d = \mathcal{E}_0 \frac{d\Phi_E}{dt}$$

$$\left(\frac{d\Phi_E}{dt}\right)_{\max} = \frac{I_{d,\max}}{\mathcal{E}_0} = 3.95 \times 10^6 \text{ V} \cdot \text{m/s}$$

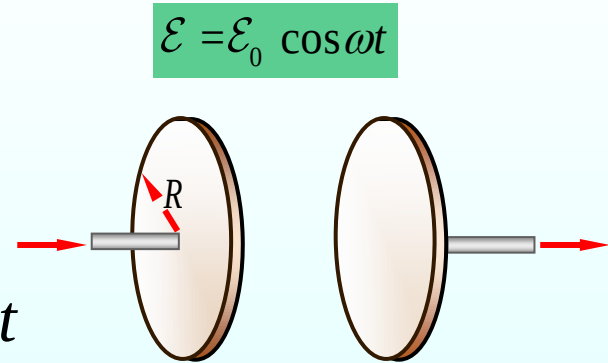
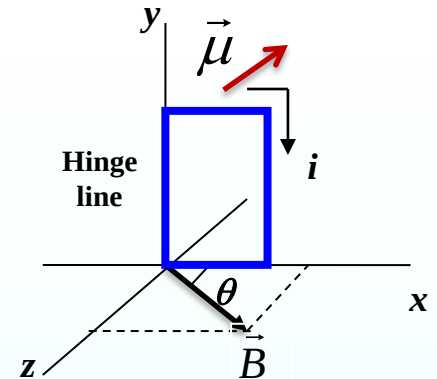


Figure shows a rectangular 20-turn coil of wire, of dimensions 10cm by 5.0cm. It carries a current of 0.10A and is hinged along one long side. It is mounted in the xy plane, at angle $\theta = 30^\circ$ to the direction of a uniform magnetic field of magnitude 0.50T. In unit-vector notation (矢量形式), what is the torque action on the coil about the hinge line?



$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

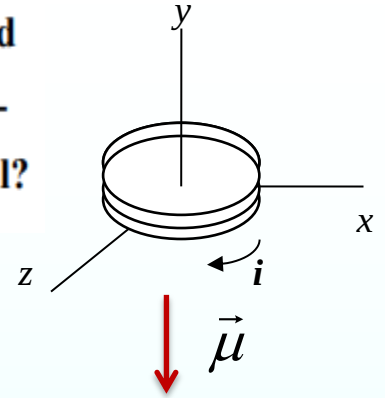
$$\vec{\mu} = NIA\vec{n}$$

$$\begin{aligned}\tau &= NiAB \sin\left(\frac{\pi}{2} + \theta\right) = NIAB \cos \theta \\ &= (20)(0.10 \text{ A})(0.10 \text{ m})(0.050 \text{ m})(0.50 \text{ T}) \cos 30^\circ \\ &= 0.0043 \text{ N} \cdot \text{m} .\end{aligned}$$

Direction : $-\hat{y}$

$$\vec{\tau} = -0.0043 \hat{j} \text{ (N} \cdot \text{m)} .$$

10-5 The coil in figure carries $i = 2.00 \text{ A}$ in the direction indicated, is parallel to an xz plane, has 3.00 turns and an area of $4.00 \times 10^{-3} \text{ m}^2$, and lies in a uniform magnetic field $\vec{B} = (2.00\vec{i} - 3.00\vec{j} - 4.00\vec{k}) \text{ mT}$. What are (a) the magnetic potential energy of the coil-magnetic field system and (b) the magnetic torque(in unit-vector notation) on the coil?



$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\vec{\mu} = NIA\vec{n} = (NiA)(-\vec{j}) = -(0.0240 \text{ A} \cdot \text{m}^2) \vec{j}$$

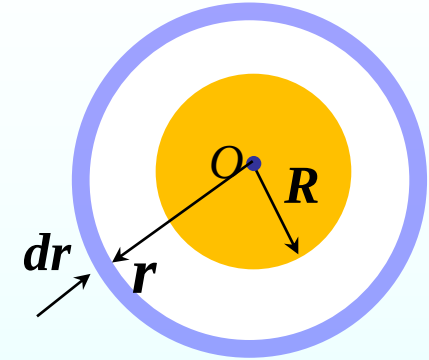
$$\begin{aligned} (a) \quad U &= -\vec{\mu} \cdot \vec{B} \\ &= -(-0.0240 \text{ A} \cdot \text{m}^2)(-3.00 \times 10^{-3} \text{ T}) = -7.20 \times 10^{-5} \text{ J} \end{aligned}$$

$$\begin{aligned} (b) \quad \vec{\tau} &= \vec{\mu} \times \vec{B} = (-0.0240 \vec{j}) \times (2.00\vec{i} - 3.00\vec{j} - 4.00\vec{k}) \times 10^{-3} \\ &= (4.8 \times 10^{-5} \vec{k} + 9.6 \times 10^{-5} \vec{i}) \text{ N} \cdot \text{m} \end{aligned}$$

Show that electrostatic energy stored in the electric field outside an isolated spherical conductor of radius R carrying a net charge Q is $Q^2 / 8\pi\epsilon_0 R$. Do this in **three** ways.

Solution 1 : In terms of electric energy density

$$E = \begin{cases} 0 & (r < R) \\ \frac{Q}{4\pi\epsilon_0 r^2} & (r > R) \end{cases}$$



$$\begin{aligned} U_e &= \int u_e dV = \int \frac{1}{2} \epsilon_0 E^2 dV \\ &= \int_R^\infty \frac{\epsilon_0}{2} \left(\frac{Q}{4\pi\epsilon_0 r^2} \right)^2 4\pi r^2 dr = \frac{Q^2}{8\pi\epsilon_0 R} \end{aligned}$$

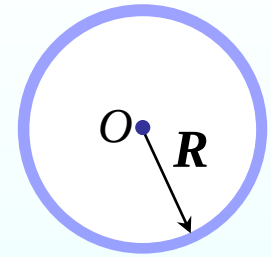
Solution 2 : Isolated conductor

$$C = 4\pi\epsilon_0 R \quad U_e = \frac{Q^2}{2C} = \frac{Q^2}{8\pi\epsilon_0 R}$$

Solution 3: Calculate the **work** needed to bring all charge **Q** up from infinity in infinitesimal bits **dq**

$$U_e = \int V dq$$

q



$$V = \frac{q}{4\pi\epsilon_0 R} \quad \text{导体球表面电势 (半径为 } R, \text{ 电量为 } q)$$

$$U_e = \int V dq = \int_0^Q \frac{q}{4\pi\epsilon_0 R} dq = \frac{Q^2}{8\pi\epsilon_0 R}$$